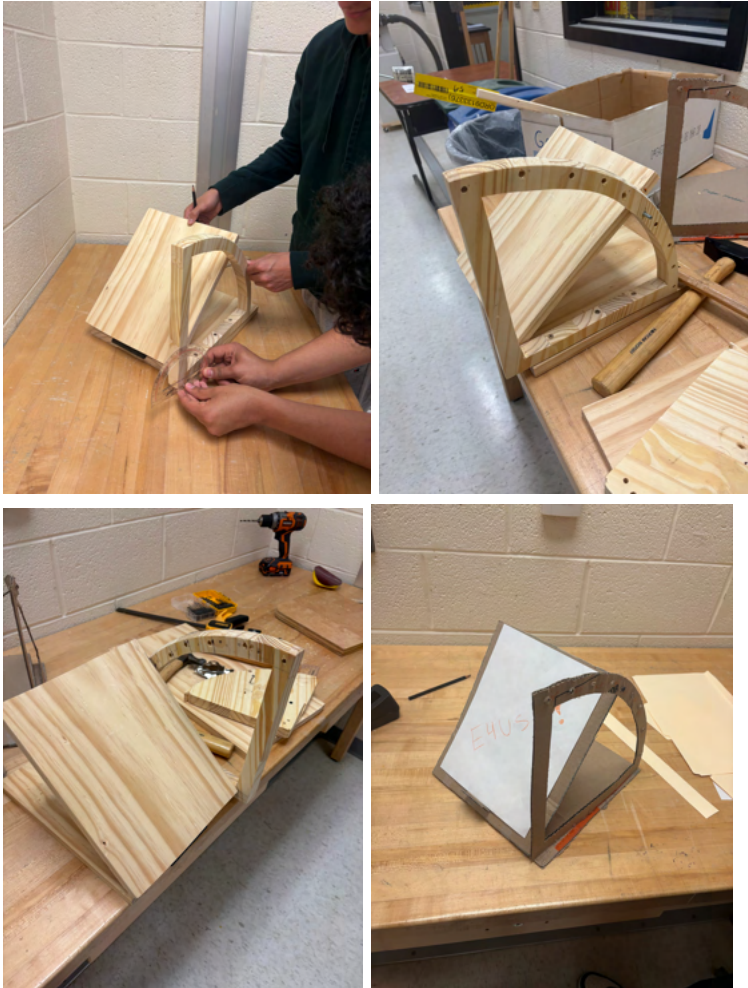


Element G	Information	Other Details	Date Added
G1	Our final iteration includes these qualities: 1. It contains two boards connected by a hinge that allows it to be slanted upwards and downwards from 0 to 90 degrees. 2. The board may be set at each angle measurement, incrementing by 10 degrees, using a “hole and pin” design, in which we can secure the board with a simple, single-user mechanism. 3. There are clear labels along the “wooden protractor” that display to the user where they can secure the board during each trial 4. The board also has a clipping mechanism that can secure the paper flat upon the wooden board and can be easily replaced between trials, “single-handedly.” 5. The Blood Splatter Apparatus is sleek and sanded so that it is both visually/physically appealing and safe for all users to operate in a timely manner. building process pictures: 		5/19/25
G2,G3,G4	Testable Requirements: Angle Adjustment Accuracy - The apparatus must allow paper to be		5/20/25

	positioned at angle increments of exactly 10 degrees, ranging from 0° to 90°, to enable accurate data collection. - The apparatus has clear markings, precisely measured and re-confirmed through multiple trials to produce a functioning mechanism that enables user accuracy constantly. Secure Paper Holding Mechanism - The apparatus must securely hold standard 8.5" x 11" paper without slipping when angled, to ensure consistency in spatter patterns. - With the addition of a large paper clip that can be easily set and/or released for the swift process of changing out the paper being used for the lab. The clip secures the paper so that it will not move beyond the initial location, allowing for precise measurements to be taken and accurate splatters of blood to be dropped. Angle Locking Feature - Each angle setting must have a locking mechanism that prevents unintentional movement once the angle is selected. - With the "hole and pin" addition, each angle may be set by inserting a metal pin through both the hole on the side of the wooden board, as well as one of the chosen wholes on the side of the "wooden protractor" depending on which angle the user wants to set the apparatus to. Easy Adjustment by Students - The angle of the apparatus must be adjustable in under 30 seconds by a high school student without tools, ensuring usability during class time. - The mechanism is easily accessible to all students and teachers, able to function with one person easily and swiftly. Each hole on the "wooden protractor" aligns almost perfectly with the hole on the side of our board, therefore making it easy for the user to insert the metal pin. Clear Angle Markings - The apparatus must have visible, labeled markings at each 10° increment to allow precise and fast adjustments during experiments. - The apparatus contains clear angle measures from 0 to 90 degrees marked with a Sharpie at the top of the arch of our "wooden protractor." These measurements were first taken using a protractor, then re-checked, and confirmed a third time to ensure the accuracy of each angle measurement. Stability on Flat Surfaces - The base of the apparatus must remain stable and unmoved on flat lab tables during use to prevent data inconsistencies. - With a flat wooden surface, the apparatus functions in a stable, sturdy manner with little to no movement by the mechanism's base, allowing for precise and unaltered evidence to be taken. Fits Within Desk Space - The entire apparatus must fit within a table space, ensuring compatibility with typical classroom lab tables. - The entire apparatus is approximately 10 inches by 14 inches, which fits within the dimensions of a standard science table/desk Cost-Effective for Classroom Use - The estimated cost per unit must		
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not exceed \$50 to ensure affordability for multiple classrooms to implement. - The total project cost amounted to \$45.50 to construct, therefore meeting the requirement, being less than the maximum cost for our project.

Non-Testable Requirements:

Durability Under Use - The materials used must withstand daily classroom use for at least one semester (approx. 1 year) without structural failure or degradation. - With strong, wooden materials being used in construction, as well as solid metal hinges securing the two boards, the apparatus should be able to survive the frequent usage by students for at least one year and likely beyond, although there is little way that we can successfully test this requirement under the small time constraint we are required to meet for this project.

Aesthetically Pleasing to the User - The labels and markings along the apparatus seem to us, the creators of such a mechanism, to be clear and straightforward, although it is hard to depict exactly, with objective and mathematically proven methods. Despite this, users may suggest additions or labels to improve its clarity, or rather, they may understand what we have currently along the sides and top of our apparatus, therefore assuring the clarity and aesthetically pleasing aspects of our mechanism.

Final Iteration Pictures:



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